

# Know about OpenShift Ingress

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## 1. What is Ingress in OpenShift v4?

**Ingress** in OpenShift v4 is a Kubernetes-native way of managing external access to services within the cluster. It allows incoming HTTP(S) traffic from outside the cluster to reach internal services. OpenShift's ingress is implemented via the **Ingress Controller**, which uses **HAProxy** as the underlying load balancer.

### Key Features of Ingress:

- Provides Layer 7 (HTTP/HTTPS) routing, Supports TLS termination
- Path-based and hostname-based routing, Load balancing for backend services

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## 2. Differences Between Ingress and Egress in OpenShift v4

| Feature          | Ingress                                  | Egress                                 |
|------------------|------------------------------------------|----------------------------------------|
| Direction        | Incoming traffic to cluster services     | Outgoing traffic from cluster services |
| Purpose          | Exposes services to external users       | Controls and monitors outgoing traffic |
| Example Use Case | Accessing a web application from outside | Restricting external communication     |
| Configuration    | Managed via Ingress/Route objects        | Managed via NetworkPolicy              |

### Commands to Check:

- **Ingress:** oc get ingress
- **Egress:** Configured through NetworkPolicy using egress rules.

### 3. Differences Between Ingress and Routes

| Feature         | Route                                           | Ingress                               |
|-----------------|-------------------------------------------------|---------------------------------------|
| Implementation  | OpenShift-specific object                       | Kubernetes-native object              |
| TLS Termination | Full, Passthrough, Edge                         | Full, Passthrough (depends on config) |
| Usage           | Easier for OpenShift-specific use               | More standardized Kubernetes-native   |
| Customizations  | Supports annotations for extended functionality | Limited compared to Routes            |

#### Command to Check Routes:

```
oc get routes
```

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### 4. Scale Ingress Pods

#### Scaling Ingress Pods:

You can scale the number of ingress pods to ensure high availability and handle increased traffic.

#### Command:

```
oc scale deployment ingresscontroller-default --replicas=<number> -n openshift-ingress-operator
```

#### Example:

```
oc scale deployment ingresscontroller-default --replicas=3 -n openshift-ingress-operator
```

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### 5. Move Ingress Pods to Infra Nodes

To ensure ingress pods only run on infrastructure nodes, use **node selectors** or **taints and tolerations**.

#### Steps:

1. Label the infrastructure nodes:

```
oc label node <node-name> node-role.kubernetes.io/infra=""
```

2. Update the ingress controller to use node selectors:

```
oc edit ingresscontroller default -n openshift-ingress-operator
```

Add the following under spec:

nodePlacement:

nodeSelector:

matchLabels:

node-role.kubernetes.io/infra: ""

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## 6. Enable Debug Logs in Ingress Controller

Enabling debug logs helps in diagnosing issues related to ingress traffic.

### Steps:

1. Edit the ingress controller to increase the log level:

```
oc edit ingresscontroller default -n openshift-ingress-operator
```

2. Add or update the logLevel under spec:

spec:

logging:

access:

enabled: true

level: debug

3. Restart the ingress pods:

```
oc delete pod -l app=router -n openshift-ingress
```

### View Logs:

```
oc logs <ingress-pod-name> -n openshift-ingress
```

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## 7. Other Important Points to Consider for OpenShift Ingress

- **Monitor ingress traffic** using tools like **OpenShift Monitoring** and **Prometheus**.
- **Use TLS encryption** for secure traffic handling.
- **Implement rate limiting** and request filtering for better control.
- **Regularly update the ingress configuration** to apply security patches.
- **Backup and document all ingress rules** for disaster recovery.

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**Commands to Check Status:**

```
oc get ingresscontroller -n openshift-ingress-operator
```

```
oc describe ingresscontroller default -n openshift-ingress-operator
```

**Please note:** In most of the Organizations, the ingress pods would be moved from default worker nodes to Infra nodes. By doing this activity, the ports 80/443 will be pointed (firewall rules) pointed to infra nodes only. Without doing the proper label and scheduling properties for infra to get the ingress pods may lead to the console, cli access unavailable.

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